WORKED SOLUTION

STEP II 2018, Question 3

Pure • Rotational Symmetry and Integrals

STEP II 2018, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s2-q3

The key idea

A half-turn about a point (a, b) sends (x, y) to $(2a - x, 2b - y)$, so a curve with that symmetry pairs each value of x with its reflection $2a - x$ and makes the two heights add to $2b$. For the tangent function the natural pairing is $x \leftrightarrow \frac{1}{2}\pi - x$, because $\tan(\frac{1}{2}\pi - x) = 1/\tan x$. That one identity gives the symmetry of the curve, and then an integral over an interval symmetric about the centre can be found without ever integrating.

Solution

Part (i)

Writing $f(x) = (1 + \tan x)^{-1}$ and using the chain rule,

$$f'(x) = -\frac{\sec^2 x}{(1+\tan x)^2} = -\frac{1}{\cos^2 x\,(1+\tan x)^2} = -\frac{1}{(\cos x + \sin x)^2}$$
$$= -\frac{1}{\cos^2 x + 2\sin x \cos x + \sin^2 x} = -\frac{1}{1+\sin 2x},$$

as required. (Multiplying $1 + \tan x$ by $\cos x$ is the step that tidies everything up.)

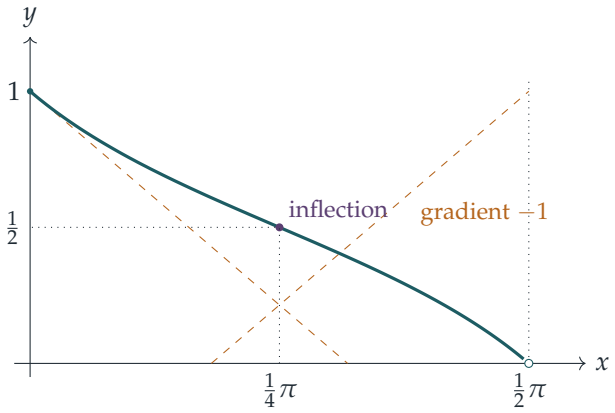
For the range, $0 \le x < \frac{1}{2}\pi$ gives $0 \le 2x < \pi$, so $\sin 2x$ takes every value in $[0, 1]$: the value 0 at $x = 0$ and the value 1 at $x = \frac{1}{4}\pi$. Hence $1 \le 1 + \sin 2x \le 2$, and taking reciprocals (which reverses the inequalities, as everything is positive) and then negatives,

$$-1 \le f'(x) \le -\tfrac{1}{2}.$$

Both ends are attained: $f'(0) = -1$ and $f'(\frac{1}{4}\pi) = -\frac{1}{2}$.

For the sketch: $f(0) = 1$, $f(\frac{1}{4}\pi) = \frac{1}{2}$, and $f(x) \to 0$ as $x \to \frac{1}{2}\pi$ because $\tan x \to \infty$. The curve is always decreasing, with gradient between -1 and $-\frac{1}{2}$. It is steepest at $x = 0$ (gradient -1), least steep at $x = \frac{1}{4}\pi$, and steepens again towards gradient -1 as

$x \to \frac{1}{2}\pi$. So f' has a maximum at $\frac{1}{4}\pi$, which is a point of inflection: the curve is concave up on $[0, \frac{1}{4}\pi)$ and concave down on $(\frac{1}{4}\pi, \frac{1}{2}\pi)$. (Indeed $f''(x) = 2\cos 2x/(1+\sin 2x)^2$, which changes sign at $\frac{1}{4}\pi$.)



The dashed lines have gradient -1: one is the tangent at $(0,1)$, and the other is the direction the curve approaches as $x \to \frac{1}{2}\pi$. The curve looks symmetric under a half-turn about $(\frac{1}{4}\pi, \frac{1}{2})$, which is what the rest of the question is about.

Part (ii)

A rotation through π about (a,b) sends (x,y) to $(2a-x, 2b-y)$, since (a,b) is the midpoint of a point and its image. The curve has rotational symmetry of order 2 about (a,b) exactly when this rotation maps the curve onto itself. (For that to be possible the domain must be symmetric about $x=a$, so that $2a-x$ is in the domain whenever x is.)

($\Rightarrow$) Suppose the curve has this symmetry. For any x in the domain, $(x, g(x))$ is on the curve, so its image $(2a-x,\, 2b-g(x))$ is on the curve. The only point of the curve with x-coordinate $2a-x$ is $(2a-x,\, g(2a-x))$, because the curve is a graph. So $g(2a-x) = 2b - g(x)$, that is,

$$g(x) + g(2a-x) = 2b.$$

($\Leftarrow$) Suppose $g(x) + g(2a-x) = 2b$ for every x in the domain. The image of a typical point $(x, g(x))$ is

$$(2a-x,\, 2b-g(x)) = (2a-x,\, g(2a-x)),$$

which is on the curve. So the rotation maps the curve into itself, and since doing it twice gives the identity, it maps the curve onto itself. Putting $x=a$ in the identity gives $g(a) = b$, consistent with (a,b) being on the curve.

For the integral, symmetry about the origin means $a = b = 0$, so $g(-x) = -g(x)$: g is an odd function. Substituting $x = -u$ in the left half,

$$\int_{-1}^{0} g(x)\,\mathrm{d}x = \int_{0}^{1} g(-u)\,\mathrm{d}u = -\int_{0}^{1} g(u)\,\mathrm{d}u,$$

so the two halves cancel and

$$\int_{-1}^{1} g(x)\,\mathrm{d}x = 0.$$

Part (iii)

Let $h(x) = \dfrac{1}{1+\tan^k x}$ for $0 < x < \frac{1}{2}\pi$. Guided by the sketch in (i), try $a = \frac{1}{4}\pi$, $b = \frac{1}{2}$. The domain $(0, \frac{1}{2}\pi)$ is symmetric about $\frac{1}{4}\pi$, and $2a - x = \frac{1}{2}\pi - x$.

Ask yourself How do we simplify $h(\frac{1}{2}\pi - x)$?

Use $\tan(\frac{1}{2}\pi - x) = \cot x = 1/\tan x$, valid for $0 < x < \frac{1}{2}\pi$, where $\tan x > 0$. Then, multiplying top and bottom by $\tan^k x$,

$$h(\tfrac{1}{2}\pi - x) = \frac{1}{1+\tan^{-k} x} = \frac{\tan^k x}{\tan^k x + 1},$$

so

$$h(x) + h(\tfrac{1}{2}\pi - x) = \frac{1+\tan^k x}{1+\tan^k x} = 1 = 2\cdot\tfrac{1}{2}.$$

The point $(\frac{1}{4}\pi, \frac{1}{2})$ is on the curve, since $\tan\frac{1}{4}\pi = 1$ gives $h(\frac{1}{4}\pi) = \frac{1}{2}$. By (ii), the curve has rotational symmetry of order 2 about $(\frac{1}{4}\pi, \frac{1}{2})$, whatever the positive constant k.

For the integral, let $I = \displaystyle\int_{\pi/6}^{\pi/3} h(x)\,\mathrm{d}x$. The interval $[\frac{1}{6}\pi, \frac{1}{3}\pi]$ is symmetric about $\frac{1}{4}\pi$, so the substitution $x = \frac{1}{2}\pi - u$ maps it to itself (with the ends swapped, which absorbs the minus sign from $\mathrm{d}x = -\mathrm{d}u$):

$$\begin{aligned} I &= \int_{\pi/6}^{\pi/3} h(\tfrac{1}{2}\pi - u)\,\mathrm{d}u = \int_{\pi/6}^{\pi/3} \big(1 - h(u)\big)\,\mathrm{d}u \\ &= \left(\tfrac{1}{3}\pi - \tfrac{1}{6}\pi\right) - I. \end{aligned}$$

Hence $2I = \frac{1}{6}\pi$, and

$$\int_{\pi/6}^{\pi/3} \frac{1}{1+\tan^k x}\,\mathrm{d}x = \frac{\pi}{12}.$$

Geometrically, the half-turn swaps the region under the curve with the region above it inside the rectangle $[\frac{1}{6}\pi, \frac{1}{3}\pi] \times [0, 1]$, so the area is half the rectangle, $\frac{1}{2}\cdot\frac{1}{6}\pi$. This is the argument of (ii) with g replaced by $h - \frac{1}{2}$.

Checks

- With $k = 1$ and $x = \frac{1}{6}\pi$: $h = 1/(1+\frac{1}{\sqrt{3}}) = \frac{1}{2}(3-\sqrt{3}) \approx 0.634$, and with $x = \frac{1}{3}\pi$: $h = 1/(1+\sqrt{3}) = \frac{1}{2}(\sqrt{3}-1) \approx 0.366$. These add to 1, as the symmetry says.

- The answer does not depend on k. As $k \to \infty$, h is close to 1 for $x < \frac{1}{4}\pi$ and close to 0 for $x > \frac{1}{4}\pi$, so the integral tends to the length of $[\frac{1}{6}\pi, \frac{1}{4}\pi]$, which is $\frac{1}{12}\pi$, in agreement.
- Numerical integration for $k = 0.3$, 1, 2, 3.7 and 10 gave 0.2617994 each time, matching $\pi/12 \approx 0.2617994$.

Common slips

- In (i), stating the range as $-1 < f'(x) \leq -\frac{1}{2}$: the value -1 is attained at $x = 0$, which is in the domain.
- In (ii), proving only one direction. "If and only if" needs both the symmetry $\Rightarrow$ identity and identity $\Rightarrow$ symmetry.
- In (iii), trying to integrate $1/(1+\tan^k x)$ directly. For general k there is no elementary antiderivative to find, and the symmetry is the whole point.

Beyond STEP

The trick in (iii), adding the integral to itself after the substitution $x \mapsto a + b - x$, is often called "King's rule". It is a special case of using a symmetry of the domain to simplify an integral, an idea that becomes central in group theory and in physics, where symmetries of a problem predict which integrals vanish (as in (ii)) or take simple values. The same reasoning evaluates $\int_0^{\pi/2} \frac{\mathrm{d}x}{1+\tan^k x} = \frac{1}{4}\pi$ for every $k > 0$. Siklos's *Advanced Problems in Mathematics* has many more integrals that yield to this kind of symmetry argument.

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